Business Requirements Document (BRD)

Logistics Analytics Dashboard | Tableau Internship Project

Project Area	Details
Dashboard Purpose	Enable decision-making across logistics revenue, delivery performance, carrier cost, warehouse load, and operational efficiency.
Data Period	2022 to 2024
Source Data	Raw and cleaned logistics Excel datasets with order, customer, carrier, warehouse, date, category, region, cost, discount, delivery, and rating fields.
Dashboard Pages	Executive Overview; Delivery Performance; Revenue & Cost Analysis; Operations & Warehouse Intelligence.
Data Model	Fact_Logistics with supporting dimensions for Date, Carrier, Category, Region, Country, Status, Priority, and Customer.

1. Business Context / Problem Statement

The logistics business handles orders across multiple countries, regions, carriers, warehouses, product categories, and drivers. Decision-makers need a centralized view to monitor revenue, delivery performance, shipping cost, warehouse load, driver workload, and customer satisfaction.

The raw dataset contains operational and data quality issues such as duplicate order IDs, missing carrier/region/warehouse values, inconsistent casing, null shipping costs, invalid customer ratings, and unclean numeric fields. Without a cleaned and structured dashboard, business teams cannot easily identify delayed deliveries, high-cost carriers, underperforming regions, discount impact, or warehouse workload imbalance.

This dashboard solves the need for a single analytics view that converts logistics data into actionable insights for revenue, delivery, cost, and operational decision-making.

2. Goal of the Dashboard

- Monitor overall logistics performance through revenue, orders, on-time rate, delivery days, and customer rating.
- Identify delivery delays by carrier, region, quarter, and delivery speed group.
- Analyze revenue, net revenue, discount impact, and shipping cost trends.
- Understand warehouse order volume, average shipment weight, and driver workload.
- Support operational decisions around carrier selection, warehouse planning, delivery improvement, and cost control.

3. Target Users / Stakeholder Personas

User Persona	Primary Need from Dashboard	Expected Usage
Operations Manager	Track order volume, warehouse load, delivery delays, and driver workload.	Daily or weekly monitoring of logistics efficiency.
Logistics Head	Compare carrier performance, late delivery rate, and shipping cost.	Strategic carrier and process decisions.
Finance / Revenue Analyst	Analyze gross revenue, net revenue, discount impact, and category performance.	Monthly revenue and cost review.
Warehouse Manager	Monitor warehouse order volume and average shipment weight.	Capacity planning and workload balancing.
Business Leadership	View executive KPIs and regional/category performance.	High-level performance review and decision-making.

4. Core Business Questions

1. What is the total revenue, total orders, average customer rating, and on-time delivery rate?
2. Which regions, countries, and product categories generate the highest revenue?
3. Which carriers have higher delivery days, lower ratings, or higher shipping costs?
4. How do gross revenue, net revenue, and discount impact vary by month and category?
5. Which warehouses and drivers are handling the highest workload, and where are capacity risks visible?

5. Product-Style KPIs to Track

KPI	Definition
Total Revenue / Gross Revenue	Sum of revenue generated before discount, calculated using quantity and unit price. Dashboard value: 487,616,442.
Net Revenue	Revenue after applying discounts. Dashboard value: 392,136,594.
Total Orders	Count of unique orders processed. Dashboard value: 4,904.
On-Time Rate %	Percentage of orders delivered within the accepted delivery timeline. Dashboard value: 47.96%.
Late Delivery Rate %	Percentage of orders exceeding the delivery-day threshold. Dashboard value: 52.04%.
Average Delivery Days	Average number of days between order date and delivery date. Dashboard value: 15.15.

Average Customer Rating	Average customer satisfaction score across logistics orders. Dashboard value: approximately 3.02.
Discount Impact %	Percentage reduction from gross revenue caused by discounts. Dashboard value: 19.43%.
Shipping Cost	Total logistics cost spent on carriers/shipping, used for carrier cost comparison over time.
Average Weight Per Order	Average shipment weight handled per order. Dashboard value: 248.8 kg.
Total Quantity Shipped	Total quantity of items shipped across all orders. Dashboard value: 485,655.
Driver Workload	Number of orders assigned to each driver.
Warehouse Order Volume	Number of orders processed by each warehouse.

6. Scope of the Dashboard

In Scope:

- Executive overview with revenue, orders, on-time rate, and customer rating.
- Delivery performance analysis by carrier, region, quarter, and delivery speed group.
- Revenue and cost analysis including gross revenue, net revenue, discount impact, and shipping cost.
- Operations and warehouse intelligence including warehouse volume, average weight, driver workload, and weight distribution.
- Filters for year, carrier, category, and other key business dimensions.
- Cleaned and modeled dataset using fact and dimension tables such as Date, Carrier, Category, Region, Country, Status, Priority, Customer, and Fact_Logistics.

Out of Scope:

- Real-time shipment tracking or live GPS monitoring.
- Predictive delivery forecasting or machine learning-based delay prediction.
- Customer-level personal profiling beyond available order/customer ID data.
- Automated carrier contract optimization.
- Integration with external logistics APIs or ERP systems.

7. Success Criteria (Measurable Outcomes)

- Business users can identify revenue, delivery, cost, and warehouse performance from one dashboard suite.
- Key KPIs such as total revenue, total orders, on-time rate, average delivery days, and customer rating are visible within the executive overview.
- Users can compare carrier performance and detect late-delivery patterns by region and quarter.

- Finance users can understand discount impact and the gap between gross and net revenue.
- Operations users can identify warehouse load imbalance and high-workload drivers.
- Cleaned data removes duplicate orders, standardizes categories/carriers/regions, and supports reliable dashboard calculations.

Data Preparation Notes

Area	Observed Issue / Treatment
Raw Data Quality	Raw data included duplicate order IDs, missing values, inconsistent casing, unclean dates/numeric values, and invalid ratings.
Cleaned Dataset	Cleaned file retains the business fields while standardizing dimensions and removing duplicate order IDs.
Modeling Approach	A star-style model supports Tableau dashboard development through a central Fact_Logistics table and dimension tables.
Dashboard Planning	Dashboard pages are organized by executive, delivery, revenue/cost, and warehouse/operations decision themes.